Jian Feng

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EMPLOYMENT

Manulife Financial Corporation

Co-op 2026 –

- Cut AKS namespace onboarding from days to minutes by automating ServiceNow tasks into Terraform/GitOps PRs with auto change tickets.
- Built a Foundry LLM agent that validates cert-SAN gitops PRs against runbooks across 9 AKS clusters, replacing manual review and catching misconfigs before prod.
- Built Power Apps-driven DL automation framework reducing manual membership updates to minutes via event-driven Graph API sync.

Treasury Board of Canada Secretariat

Co-op 2025

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Lansi Institute of Artificial Intelligence in Medicine

Co-op (Remote) 2024

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd.

Intern 2024

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

EDUCATION

University of Waterloo 2023-2028

Bachelor of Computing and Financial Management

- Majors: Computer Science, Finance, Statistics
- Grew UX Research Hub from 10→318 members as Director; secured Google UXR keynote at $0 club cost and TikTok chapter cross-promotion driving sustained inbound growth.
- Velocity Co-op Problem Award, 3rd in 12th SAF Investment Research Challenge, Finalist in 7th SAF Strategy Case Competition
- Certifications: Duke Data Engineering, IBM Vector Database, UCI Database Design & Operational BI, IBM AI for Cybersecurity, Stanford Machine Learning, UCI Mathematics for ML, UCSC Bayesian Statistics, Rice Parallel-Concurrent-Distributed Programming, UA Reinforcement Learning

PERSONAL PROJECTS

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

GhostPilot

- Invisible Windows overlay streams WASAPI loopback audio through partial/final Azure ASR into a text LLM.
- Alt+P vision path: Qt area capture → JPEG → vision LLM, result folded back into the text-LLM prompt.
- Local RAG over `knowledge/` injects user-curated context so answers stay grounded across long sessions.

MiniKV

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

Anchor

- Splunk drift agent fingerprints a healthy window, then ranks diffs against later windows with LLM-narrated SPL suggestions.
- Feedback loop reweights signals and surfaces recurring unresolved drifts as blind spots.

ChefCopilot

- iOS SwiftUI monorepo: on-device CV → SwiftData inventory → FastAPI recipe ranker (feasibility + diversity) → cook-mode step guidance.

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

TECHNOLGY

Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, React, Node.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, PyTorch, TensorFlow, Scikit-learn, OpenCV, pandas, Microsoft Agent Framework